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CO2_AT1_MCQ TEST

Total points 20/20 ?

NAME

Vidhya sagar.E

REGISTER NO

192565024

- ✓ In informed search strategies, some algorithms use only the heuristic value to decide which node to expand next. This approach focuses on selecting the node that appears closest to the goal without considering the cost already incurred. 1/1
- Which algorithm follows this principle?

☐ Breadth First Search

☐ Uniform Cost Search



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- ☐ Breadth First Search
- ☐ Uniform Cost Search
- ☒ Greedy Best-First Search ✓
- ☐ Depth First Search



✓ A* search is widely used in

1/1

☐ Uniform Cost Search☒ Greedy Best-First Search ✓☐ Depth First Search

✓ A* search is widely used in pathfinding and graph traversal because it combines the actual cost from the start node and the estimated cost to the goal. This balance allows it to find optimal solutions efficiently when the heuristic is admissible. What is the evaluation function used in A* search? 1/1

☐ $f(n) = g(n)$ ☐ $f(n) = h(n)$ ☒ $f(n) = g(n) + h(n)$ ✓☐ $f(n) = g(n) \times h(n)$

✓ Heuristic functions play a crucial role in informed search by guiding the algorithm toward the goal. A good heuristic should not overestimate the actual cost and should improve search efficiency. 1/1



- ✓ Heuristic functions play a crucial role^{1/1} in informed search by guiding the algorithm toward the goal. A good heuristic should not overestimate the actual cost and should improve search efficiency.

What property ensures optimality in A* search?

☐ Completeness

☒ Admissibility



☐ Depth limit

☐ Backtracking

- ✓ Local search algorithms operate by^{1/1} starting with an initial solution and iteratively improving it based on a given objective function. These methods are commonly used in optimization problems where the path is not important but the final solution is.

Which of the following is a local search algorithm?





☐ Backtracking

✓ Local search algorithms operate by starting with an initial solution and iteratively improving it based on a given objective function. These methods are commonly used in optimization problems where the path is not important but the final solution is. 1/1

Which of the following is a local search algorithm?

☐ A* Search

☒ Hill Climbing ✓

☐ Breadth First Search

☐ Uniform Cost Search

✓ In adversarial search, agents compete against each other in a game environment. The decision-making process involves considering both the agent's moves and the opponent's possible responses. 1/1



☐ Breadth First Search

☐ Uniform Cost Search

✓ In adversarial search, agents compete against each other in a game environment. The decision-making process involves considering both the agent's moves and the opponent's possible responses. 1/1

Which algorithm is commonly used for such decision-making?

☐ Greedy Search

☒ Minimax Algorithm ✓

☐ Backtracking

☐ Uniform Search

✓ A delivery robot in a warehouse selects paths based only on the estimated distance to the destination without considering obstacles already encountered. This sometimes leads to suboptimal paths but faster decisions. 1/1

Which algorithm is the robot using?



Uniform Search



A delivery robot in a warehouse selects paths based only on the estimated distance to the destination without considering obstacles already encountered. This sometimes leads to suboptimal paths but faster decisions.

1/1

Which algorithm is the robot using?



A* Search



Greedy Best-First Search



Uniform Cost Search



Depth Limited Search



A GPS navigation system calculates routes by combining the distance already travelled and an estimate of the remaining distance to the destination. This helps in finding the shortest path efficiently.

1/1

Which algorithm best represents this approach?



✓ A GPS navigation system calculates routes by combining the distance already travelled and an estimate of the remaining distance to the destination. This helps in finding the shortest path efficiently. 1/1

Which algorithm best represents this approach?

☐ Depth First Search

☒ A* Search ✓

☐ Hill Climbing

☐ Backtracking

✓ A university is scheduling exams for multiple subjects with constraints such as no overlapping exams and limited rooms. The solution must satisfy all constraints while assigning time slots. 1/1

This problem is best modeled as:

☐ Game Playing Problem

☒ Constraint Satisfaction Problem ✓



☐ Game Playing Problem

☒ Constraint Satisfaction Problem



☐ Heuristic Search Problem

☐ Optimization Problem



✓ An AI-based chess program evaluates all possible moves and anticipates the opponent's responses before making a decision. It assigns utility values to game states to determine the best move. Which algorithm is being applied here? 1/1

☐ Greedy Search

☐ Hill Climbing

☒ Minimax Algorithm



☐ Breadth First Search

✓ A robot explores an unknown environment where it does not have prior knowledge of obstacles. It makes decisions step-by-step and updates its knowledge as it moves. 1/1





Minimax Algorithm



Breadth First Search



A robot explores an unknown environment where it does not have prior knowledge of obstacles. It makes decisions step-by-step and updates its knowledge as it moves. Which type of agent is suitable for this scenario?

1/1



Offline Agent



Online Search Agent



Static Agent



Deterministic Agent



A* search guarantees optimal solutions only when certain conditions are satisfied by the heuristic function. If these conditions are violated, the algorithm may still find a solution but not necessarily the best one.

1/1



Why is an admissible heuristic important?

- ☐ Static Agent
- ☐ Deterministic Agent

✓ A* search guarantees optimal solutions only when certain conditions are satisfied by the heuristic function. If these conditions are violated, the algorithm may still find a solution but not necessarily the best one. Why is an admissible heuristic important? 1/1

- ☐ It reduces memory usage
- ☒ It ensures optimality ✓
- ☐ It increases branching factor
- ☐ It avoids loops

✓ In large game trees, evaluating every node becomes computationally expensive. Alpha-Beta pruning helps eliminate branches that do not affect the final decision, thereby improving efficiency. What is the primary benefit of Alpha-Beta pruning? 1/1



✓ In large game trees, evaluating every 1/1 node becomes computationally expensive. Alpha-Beta pruning helps eliminate branches that do not affect the final decision, thereby improving efficiency.

What is the primary benefit of Alpha-Beta pruning?

- ☐ Reduces depth of tree
- ☒ Reduces number of nodes evaluated ✓
- ☐ Changes game rules
- ☐ Improves heuristic accuracy

✓ Hill Climbing algorithm continuously 1/1 moves toward better states based on local improvements. However, it may fail to reach the global optimum due to certain limitations in the search space.

What is the main drawback of Hill Climbing?

- ☐ High memory usage
- ☐ Revisiting nodes





Improves heuristic accuracy



Hill Climbing algorithm continuously moves toward better states based on local improvements. However, it may fail to reach the global optimum due to certain limitations in the search space.

What is the main drawback of Hill Climbing?



High memory usage



Revisiting nodes



Getting stuck in local maxima



Requires complete tree



Constraint Satisfaction Problems involve variables, domains, and constraints. A valid solution must assign values to all variables such that no constraints are violated.

What ensures a valid CSP solution?



Heuristic function